

Digital forensics is the process of collecting, analyzing, and preserving digital evidence from computers, mobile devices, networks, and cloud storage systems for legal or security purposes.

Practical applications of digital surveying:

- Tracking hackers in Advanced Persistent Threat (APT) attacks (ransomware, phishing).
- Verifying and recovering data in cases of information leaks.
- Assisting law enforcement agencies in high-tech cases.
- Detecting an insider threat – an employee leaking internal data.

Areas within digital forensics

Digital forensics focuses not only on personal computers, but also extends to networks, mobile devices, malware, and cloud platforms.

1. Computer Forensics (Computer System Investigation)

Computer forensics is a core field within digital investigation, specializing in the collection, recovery, and analysis of data from personal computers, servers, hard drives, and other storage devices.

Key investigative techniques:

- Extracting and recovering deleted data: When a file is deleted, it doesn't actually disappear immediately. Tools like Autopsy, FTK Imager, and TestDisk can help recover files from sectors that haven't been overwritten.
- Analyzing file system and metadata:
 - Search for traces of file modifications: Compare timestamps (MAC times – Modified, Accessed, Created).
 - Analyze the Windows registry to find evidence of malicious programs that have ever run.
- RAM (Memory Forensics) Examination:
 - Data in RAM is not stored permanently, but it contains important information such as keyloggers, session logins, or malware running in the background.
 - The Volatility tool, Rekall, can help analyze memory and retrieve important information.
- Tracing hackers through system logs (Windows/Linux):
 - Windows: Event Viewer, Sysmon, Security Logs.
 - Linux: /var/log/auth.log, /var/log/syslog.

Challenge:

- Data encryption (BitLocker, VeraCrypt) makes investigation difficult.
- Anti-Forensics attacks allow hackers to overwrite or delete system logs to erase their tracks.

Tools used:

- Autopsy, FTK Imager, Volatility, Rekall, Sleuth Kit.

2. Network Forensics (Network Traffic Investigation)

Network forensics focuses on analyzing data traffic moving through a network system to detect incidents, cyberattacks, cyber espionage or unusual behavior.

Key investigative techniques:

Packet Capture – Capture and analyze packets:

- Tools like Wireshark, Zeek (Bro IDS), and tcpdump help extract data from protocols such as HTTP, DNS, FTP, SSH.
- For example, if a hacker uses DNS tunneling, we can check for unusual DNS queries to trace them.

Firewall, IDS/IPS log analysis:

- Logging from Cisco ASA, Palo Alto, and FortiGate helps detect unauthorized activity.
- Check Syslog and NetFlow to track the hacker's IP address.

Investigating DDoS attacks, Man-in-the-Middle attacks, and DNS spoofing:

- If the system is DDoS, check your firewall logs to find the IP addresses of the attack sources.
- If you suspect Man-in-the-Middle, check your SSL certificates and ARP poisoning.

Challenge:

- Large amounts of network data are needed for analysis, requiring automation tools.
- Fileless attacks – Hackers can use legitimate tools to conceal their activity.

Tools used:

Wireshark, Zeek (Bro IDS), Splunk, Snort, Suricata.

3. Mobile Forensics

Mobile forensics focuses on extracting data from mobile phones, GPS, messaging apps, and smartphone storage systems.

Key investigative techniques:

Extract data from a locked or erased phone:

- Use Cellebrite UFED and Oxygen Forensic Suite to bypass the encryption key.
- Analyze the SQLite database to find messages and call history.

Analyze metadata from photos, messages, and calls to reconstruct user behavior:

- GPS location data in EXIF images can reveal the user's actual location.
- WhatsApp, Signal, and Telegram browsing history can be accessed if it is not fully encrypted.

Challenge:

- The strong encryption features on iOS and Android make data extraction difficult.
- End-to-end encrypted messaging applications (Signal, Telegram) are virtually impossible to decrypt.

Tools used:

Cellebrite UFED, Magnet AXIOM, Oxygen Forensic Suite.

4. Malware Forensics (Malware Investigation)

Malware analysis helps to understand how malware works. malware, ransomware, rootkits to deploy countermeasures.

Key investigative techniques:

- Static Analysis – Malware source code analysis: Using IDA Pro, Ghidra, and Radare2 to examine malicious API functions.
- Dynamic Analysis – Testing malware in a sandbox environment: Use Cuckoo Sandbox to test malware behavior without affecting the real system.

Challenge:

Hackers can use anti-analysis (anti-VM) techniques.

Tools used:

IDA Pro, Ghidra, Cuckoo Sandbox, YARA.

5. Cloud Forensics

Cloud Forensics focuses on analyzing logs from cloud services such as AWS, Azure, and Google Cloud to detect fraud or attacks.

Key investigative techniques:

- Extract and analyze logs from AWS CloudTrail and Google Workspace Logs.
- Check for suspicious activity such as unusual logins or changes to API permissions.

Challenge:

Data in the cloud cannot be accessed directly like on traditional computers.

Tools used:

AWS CloudTrail, Azure Security Center, Google Cloud Logging.

Digital investigation process

1. Evidence Acquisition

- Preserve data integrity by creating a forensic image.
- Use tools like FTK Imager, Autopsy, or dd to create bit-by-bit copies.
- Hashing data (MD5, SHA-256) ensures that the data remains unchanged.

```
bash CopyEdit dd if=/dev/sda of=/mnt/forensics/image.dd bs=4M md5sum /mnt/forensics/image.dd
```

2. Evidence analysis (this can be considered the most important part)

- File system analysis: Identify deleted files, metadata, and timeline events.
- RAM Forensics: Extracts malware that runs in memory.
- Network traffic analysis: Check the captured packets to trace the hacker.

For example: Analyzing RAM to find keyloggers or malware running in the background.

Use Volatility to check for suspicious processes in RAM:

```
bash CopyEdit volatility -f memory.dmp --profile=Win7SP1x64 pslist
```

If you see unusual processes running in the background, it could be a sign of malware.

3. Reporting and using evidence before the law.

- Write a detailed report on the investigation findings.
- Present evidence in a legally sound manner to ensure it is usable in court.